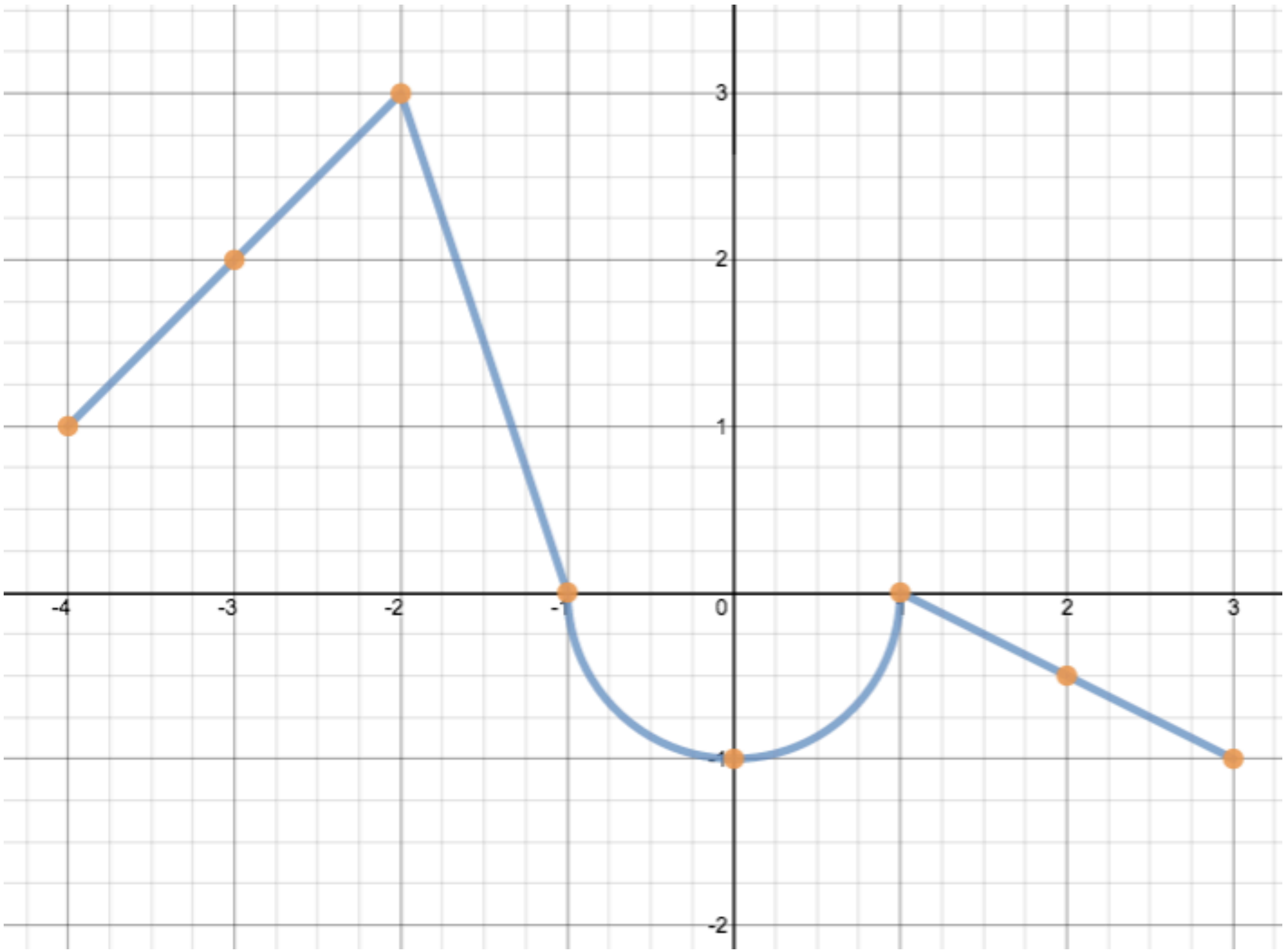
# Splines
# and
# Generalized Additive Models
# (GAMs)

Shaina Race, PhD

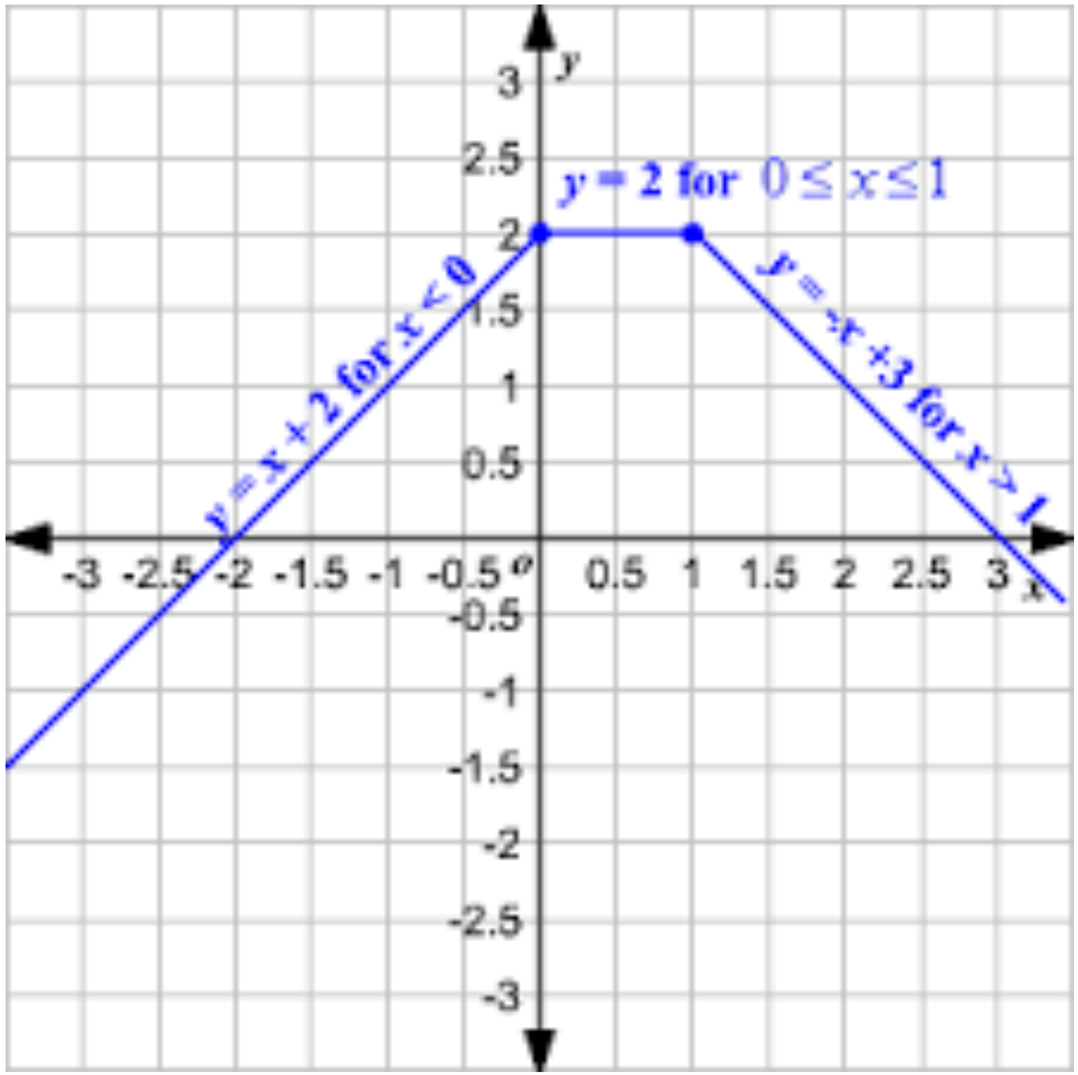
# Multivariate Adaptive Regression Splines (MARs)

Friedman (1991)

# Piecewise functions

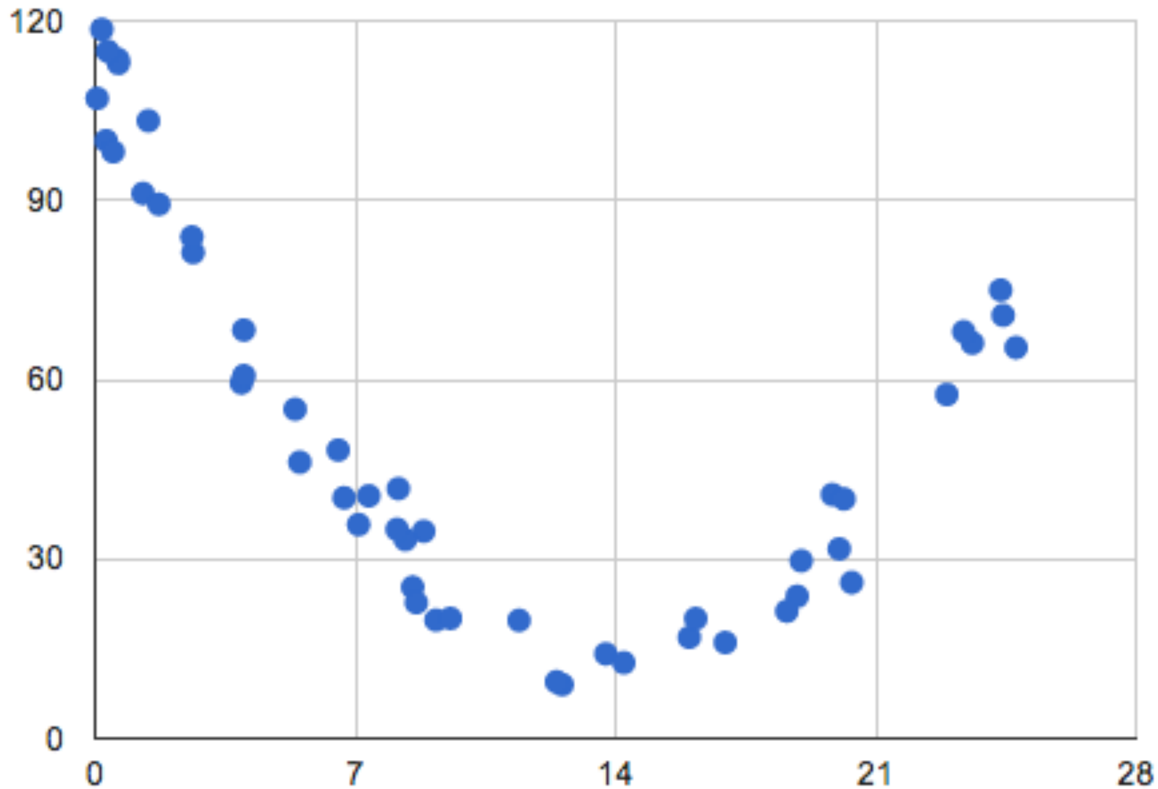


# Piecewise functions



# Knots

- MARS first looks for the point in the range of x where two linear functions on either side of the point provides the least error.



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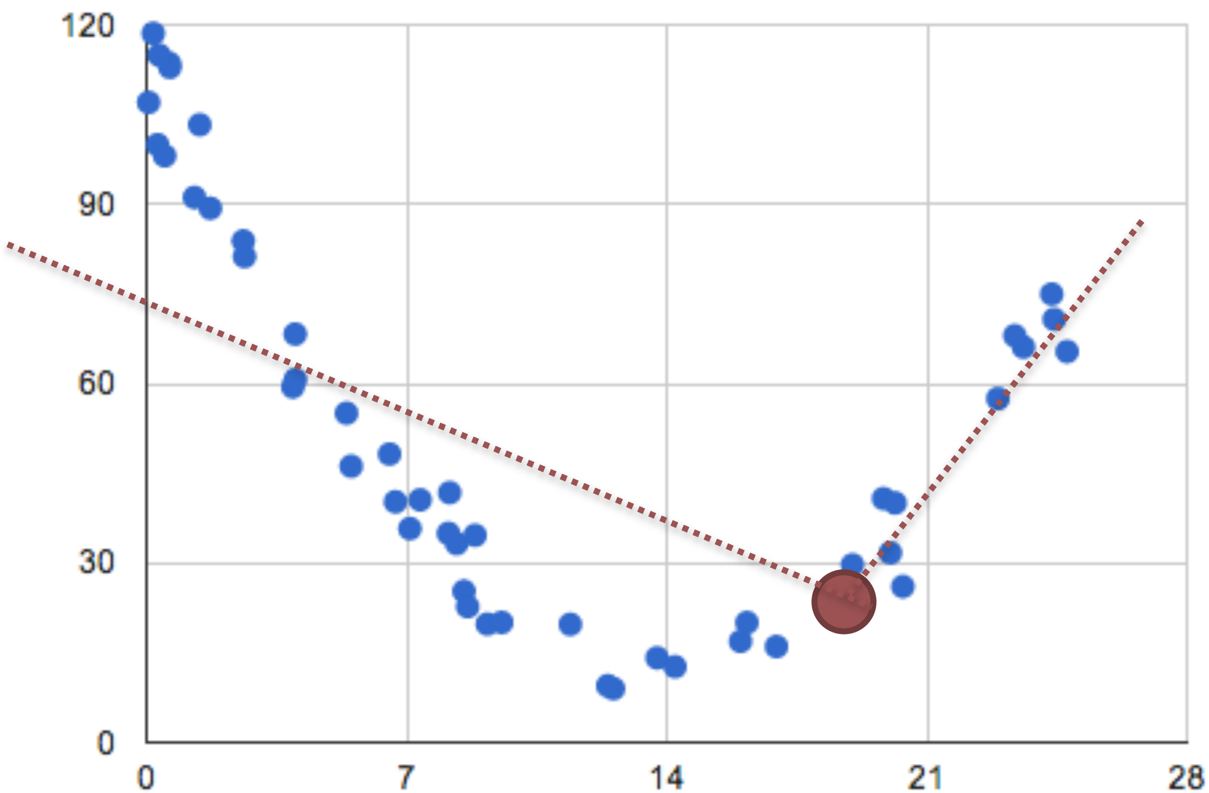

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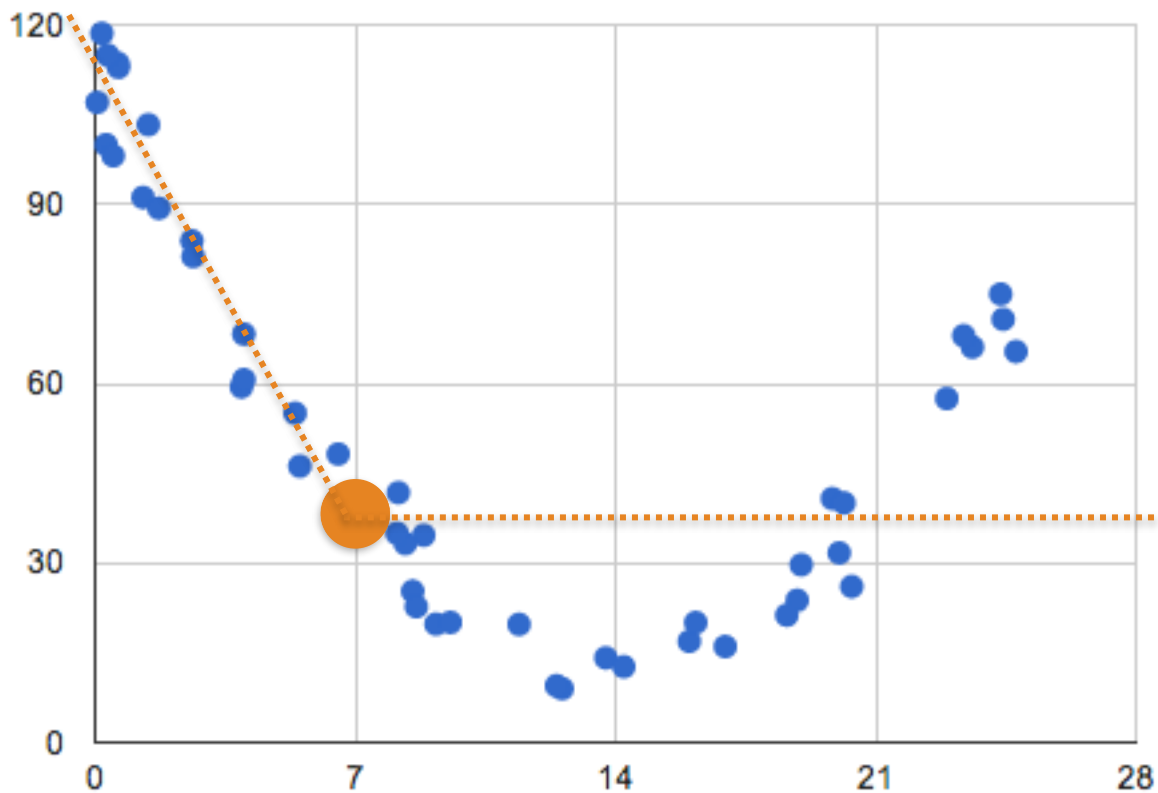

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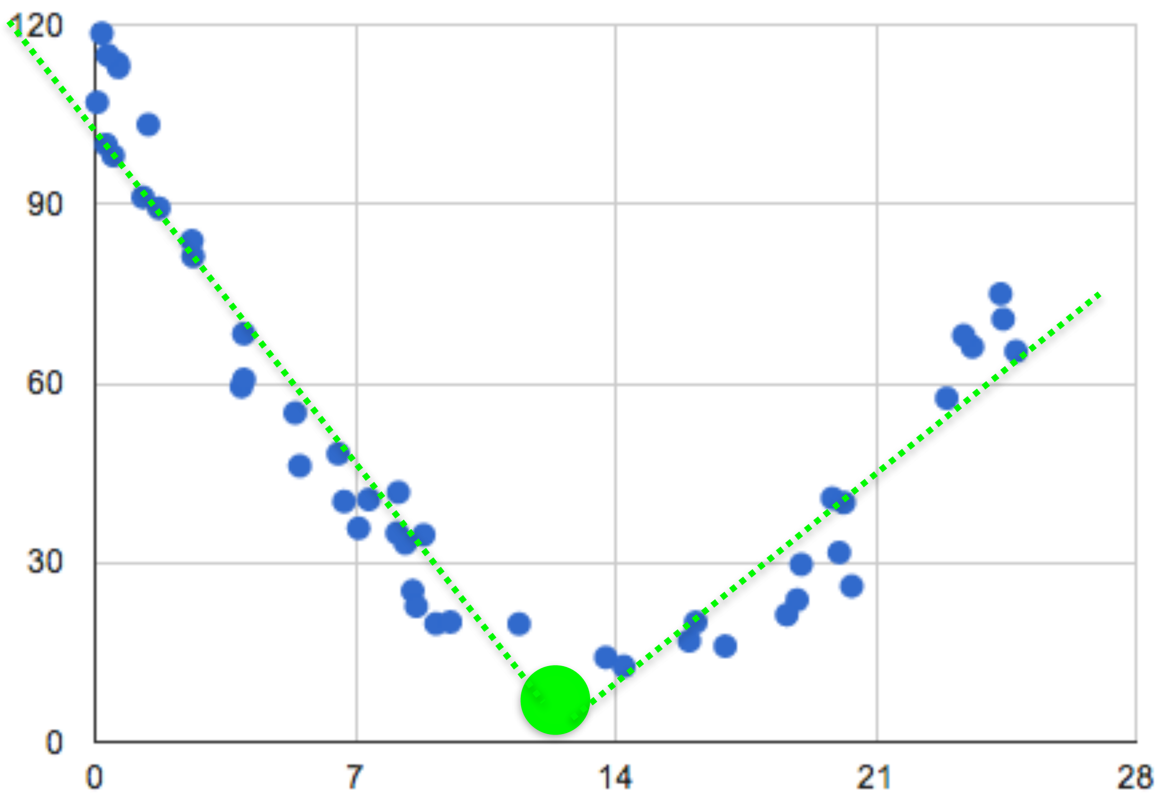

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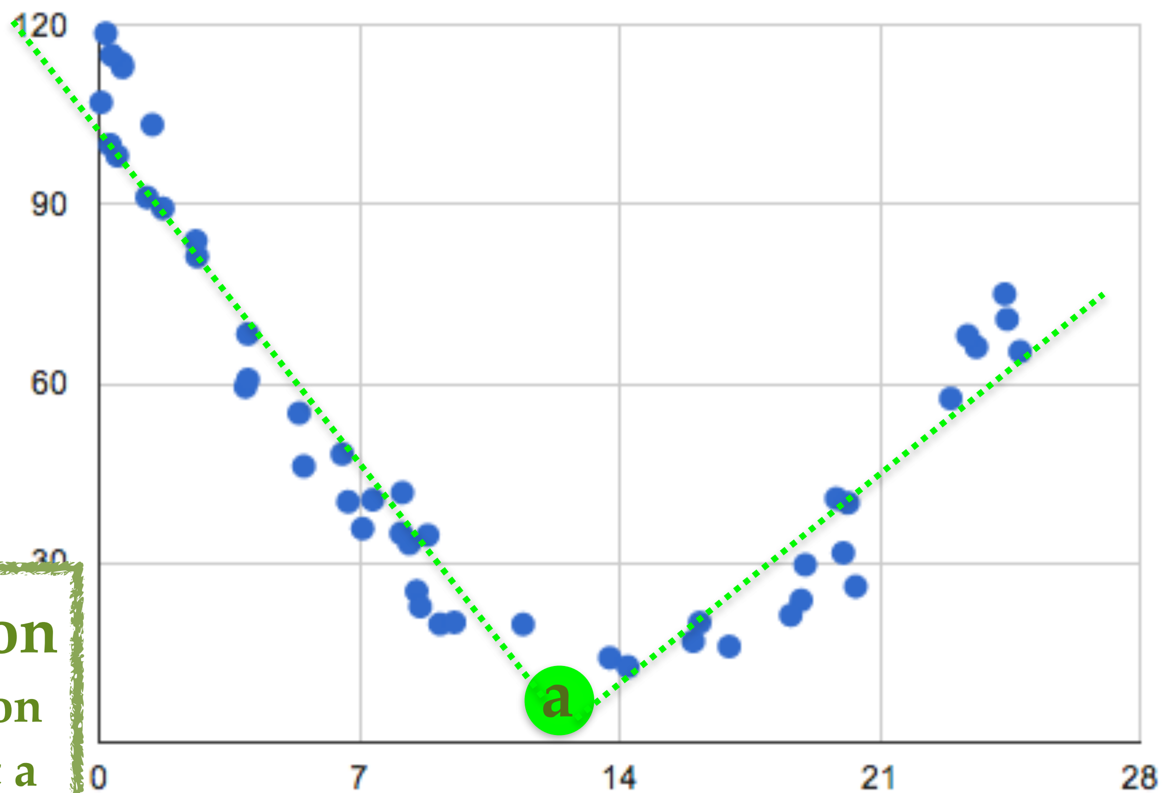
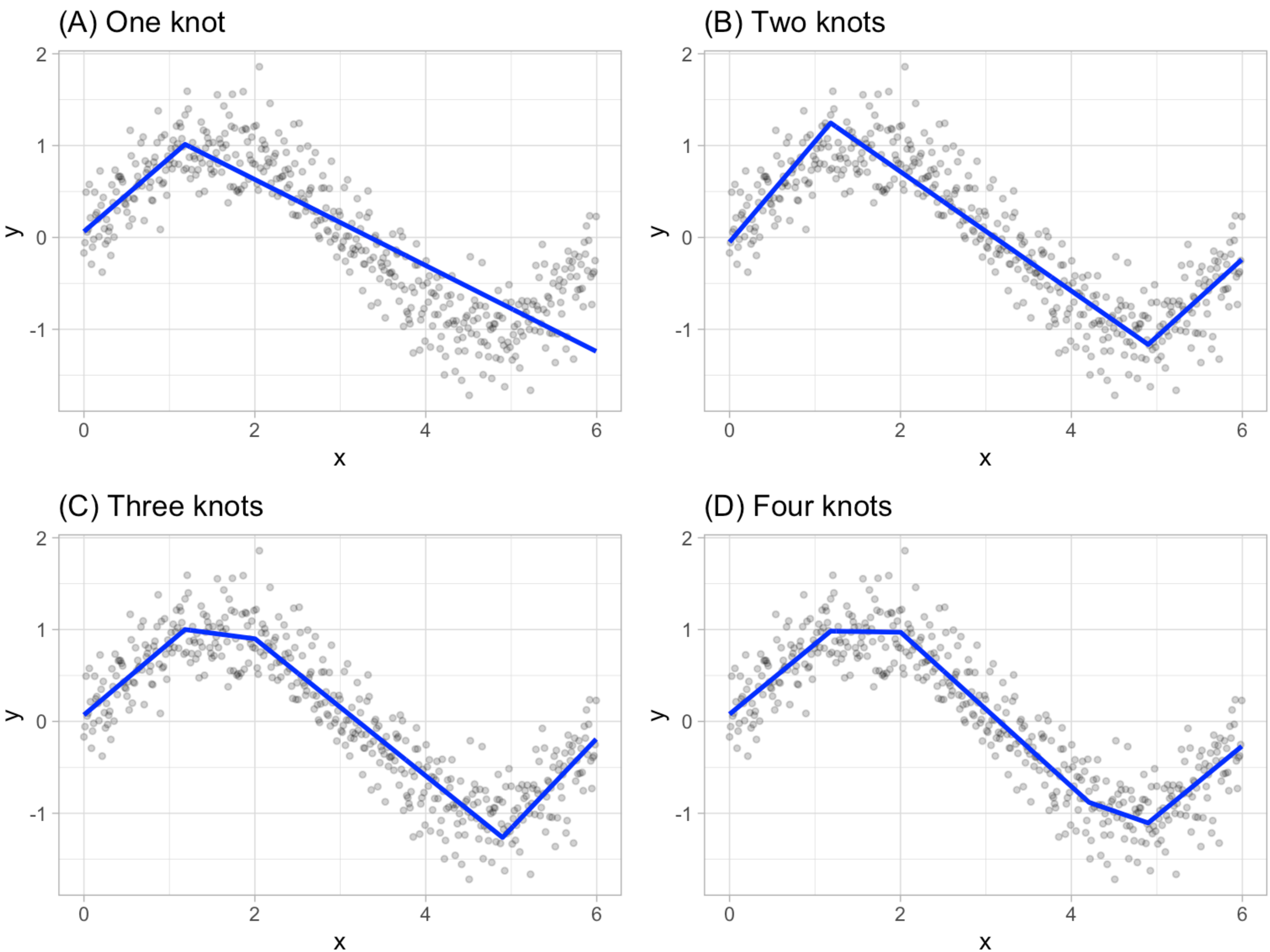



**Hinge function**

**piecewise function centered at point a**

# Knots



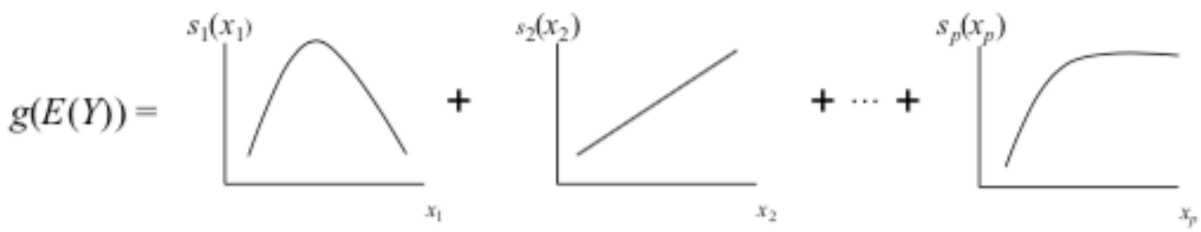
# Knots

- This procedure continuous on each piece until many knots are found
- Too many knots create too much flexibility and will overfit the training data
- Method then "prunes" the knots to remove those that do not contribute significantly to out-of-sample accuracy
- Cross-Validation typically used to determine optimal number of knots.

# General Formula of a GAM

$$y = \sum_{j=1}^{d} \beta_j g_j(x) + \epsilon$$

- $g_j(x)$ is some function of x - called a **basis function**
- $\beta_j$ is the regression coefficient



# Polynomial Regression as a GAM

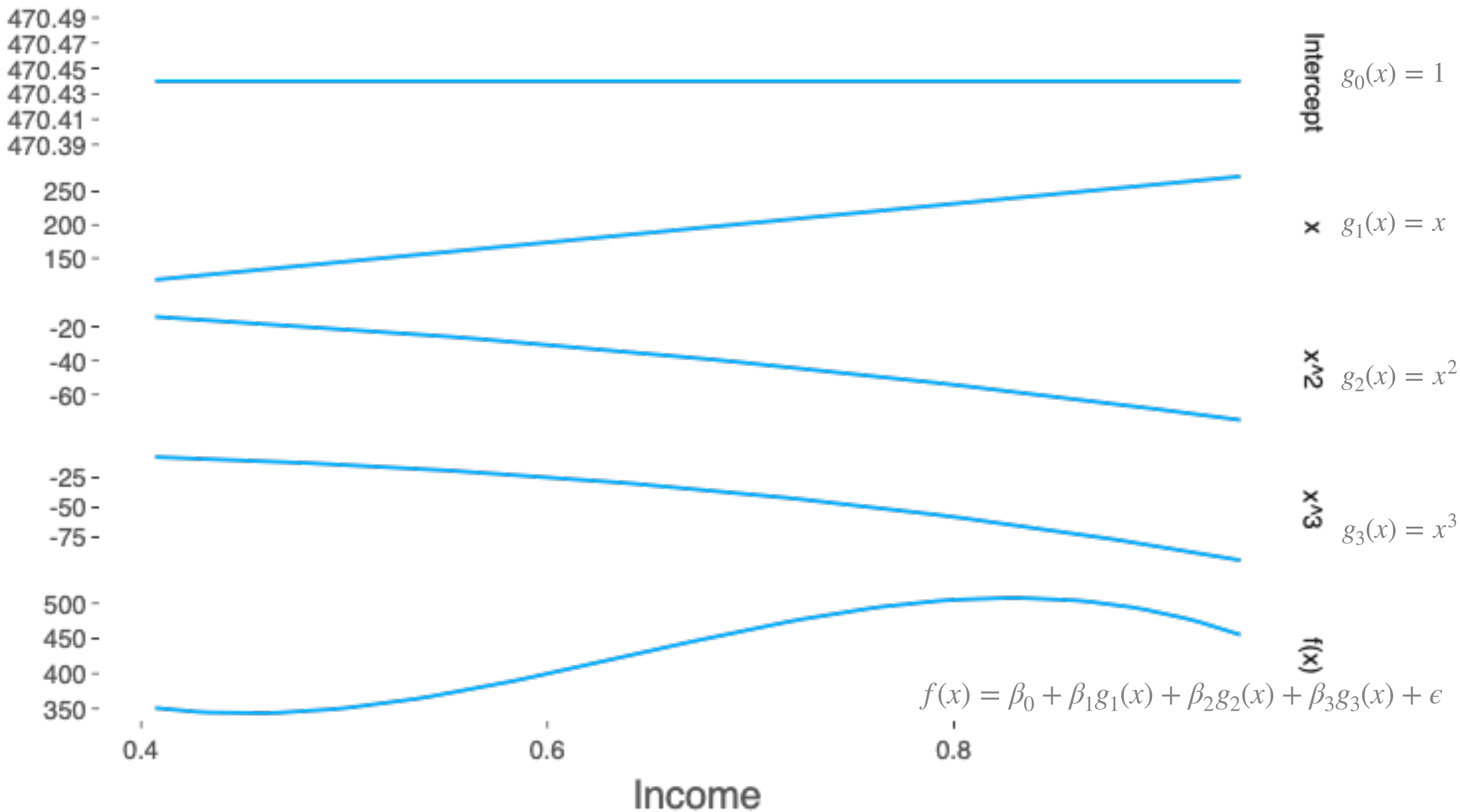
- Consider 3rd degree polynomial regression of y on x:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \epsilon$$

- Let $g_0(x) = 1$, $g_1(x) = x$, $g_2(x) = x^2$, $g_3(x) = x^3$.

- Polynomial regression fits the functional form of a GAM

# Polynomial Regression as a GAM



# Enhanced Adaptive Regression Through Hinges

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(EARTH)

Trevor Hastie and Thomas Lumley 2019

MARS acronym is trademarked and licensed to Salford Systems. The EARTH package provides an implementation of MARS.

# EARTH

By default, EARTH will use linear functions and check all possible knots across all input features and then prune back those knots based on expected change in $R^2$ of 0.001

```
load("concrete.RData")
install.packages('earth')
library(earth)
mars1 = earth(strength ~ .,  data = concrete)
print(mars1)
```

```
> print(mars1)
Selected 17 of 18 terms, and 7 of 8 predictors
Termination condition: Reached nk 21
Importance: age, cement, water, slag, ash, fineagg, superplastic, coarseagg-unused
Number of terms at each degree of interaction: 1 16 (additive model)
GCV 38.81178    RSS 37455.59    GRSq 0.8610655    RSq 0.8695723
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listed in order of importance

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Number of terms at each degree of interaction: 1 16 (additive model)
GCV 38.81178   RSS 37455.59   GRSq 0.8610655   RSq 0.8695723   CVRSq 0.8549731
> mean((mars1$fitted.values-concrete$strength)^2)
[1] 36.36465
```

**Training MSE comparable to Neural Network!**

# Variable Importance

```
> evimp(mars1)
            nsubsets   gcv    rss
age               16 100.0  100.0
cement            15  78.1   78.2
water             14  57.7   57.9
slag              13  48.2   48.6
ash               12  33.7   34.3
fineagg            8  20.3   21.1
superplastic       7  17.8   18.6
```

**There is one "subset" for each model size, and that subset is the best set of terms for that model size. Model size refers to the number of terms in the model.** Here we only consider the models that are smaller than or same sized as final model. **Our final model had 17 terms. Age was present in all models smaller than final model.** Superplastic was present in only 7 of the 16 smaller models.
*Note: this measures importance using only training data.*

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```

**RSS calculates the decrease in residual sums of squares for each subset relative to the previous subset. Then for each variable, it sums these decreases over all subsets that include the variable.** Finally the numbers are scaled so the largest is equal to 100.

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# Variable Importance

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            nsubsets   gcv    rss
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```

**The GCV approximates the RSS that would be measured on out-of-sample data (specifically, it estimates error from leave-one-out cross validation). Even when the approximation is not that good, it is usually good enough for comparing models. See page 279 in ISL for brief discussion**

# Including Interactions

The degree = option in earth specifies the maximum degree of interaction between hinge functions

```
mars1 = earth(strength ~ .,  data = concrete, degree=2)
print(mars1)
mean((mars1$fitted.values-concrete$strength)^2)
```

```
Selected 19 of 20 terms, and 7 of 8 predictors
Termination condition: Reached nk 21
Importance: age, cement, water, slag, ash, fineagg, superplastic, coarseagg-unused
Number of terms at each degree of interaction: 1 13 5
GCV 38.35217   RSS 36053.13   GRSq 0.8627107   RSq 0.874456   CVRSq 0.8558624
> mean((mars1$fitted.values-concrete$strength)^2)
[1] 35.00304
```

# Yes, classification too:

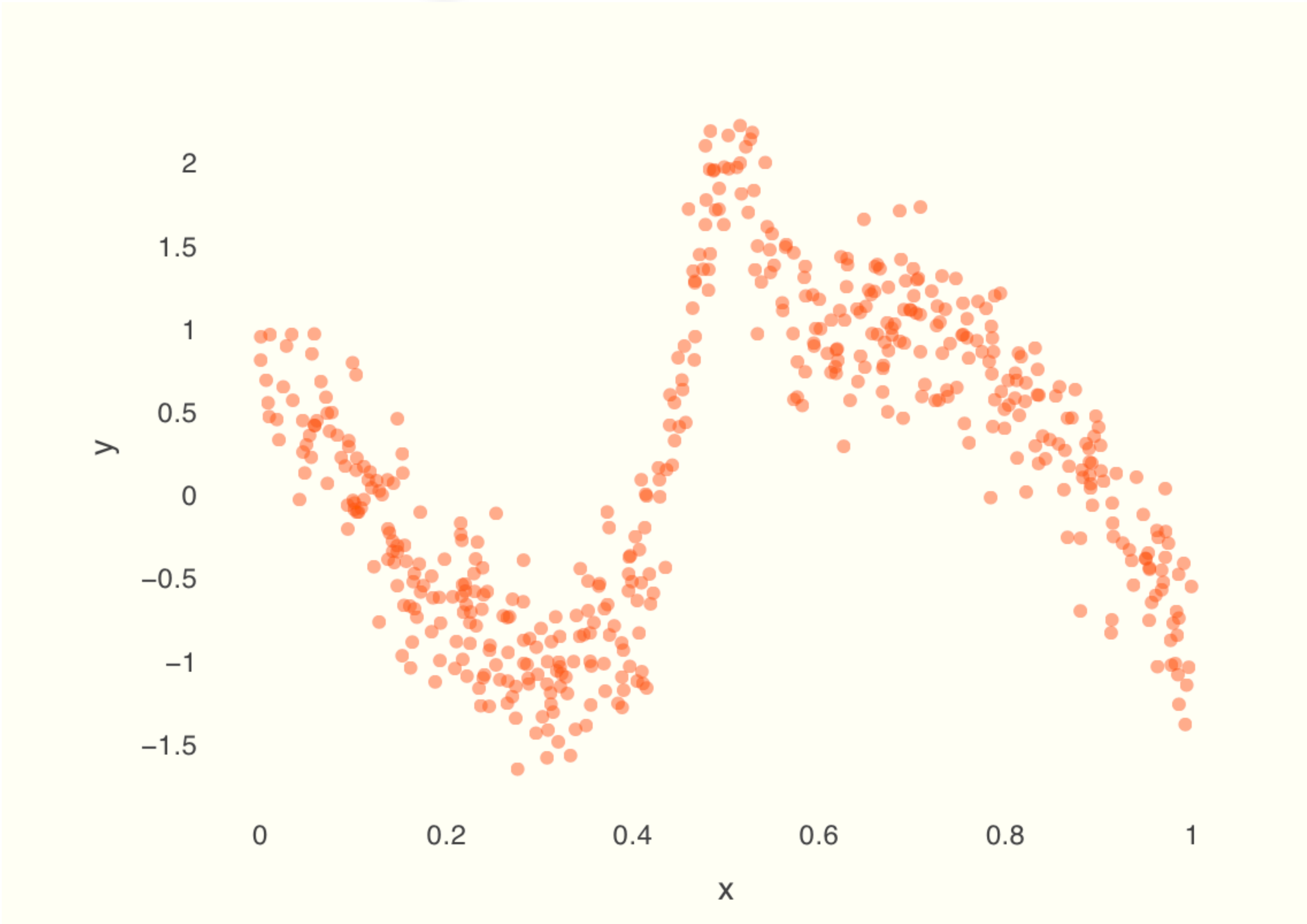
```
load("churn.Rdata")
mars2 = earth(churn~.-area_code, data = churnTrain,
              glm=list(family=binomial), degree=2)
print(mars2)
#Training Accuracy
mean(((mars2$fitted.values<0.5) == (churnTrain$churn=='yes')))
pred=predict(mars2, churnTest, type = 'class')
#Test Accuracy
mean(pred == churnTest$churn)
```

```
> mean(((mars2$fitted.values<0.5) == (churnTrain$churn=='yes')))
[1] 0.9588959
```

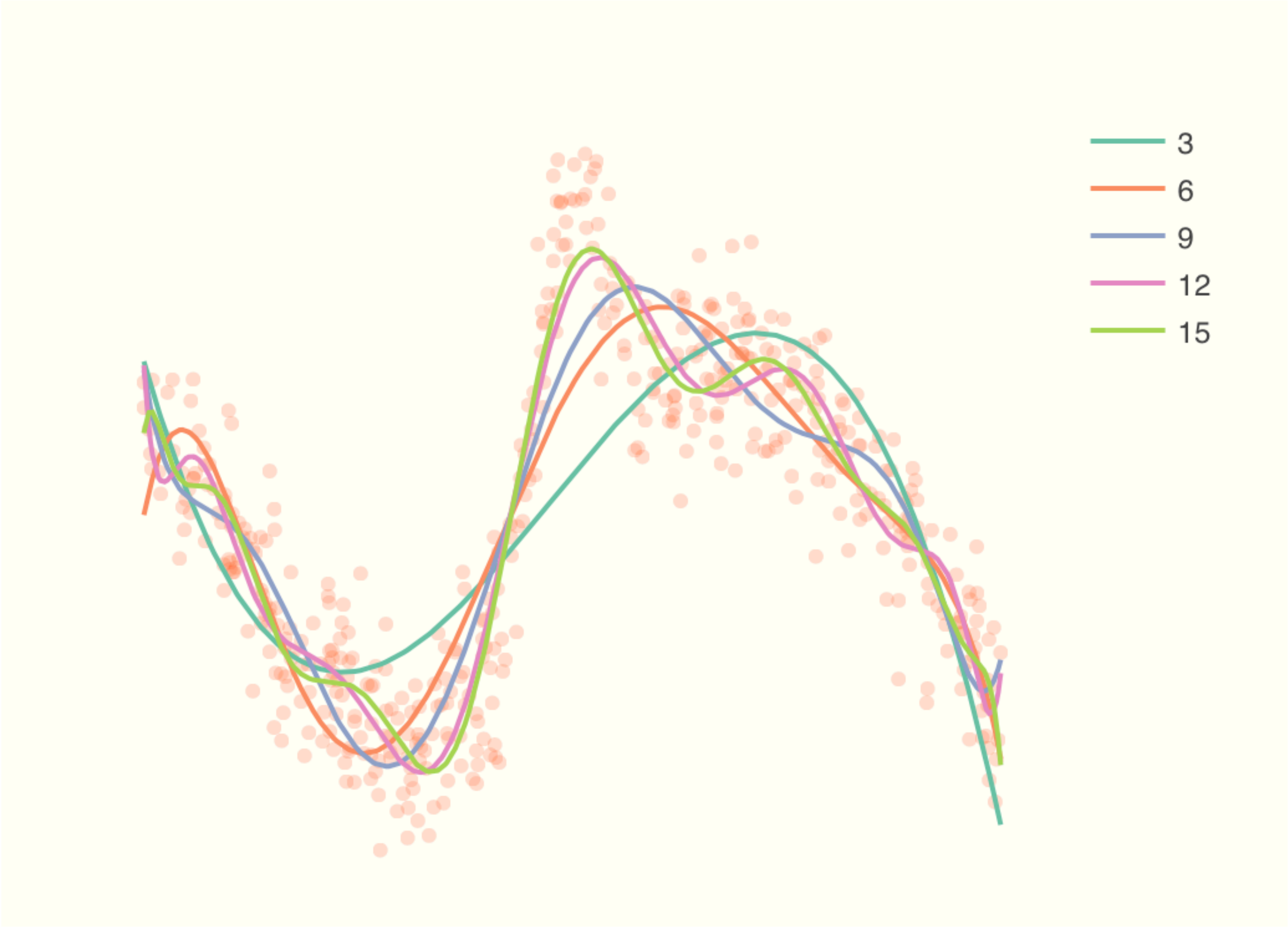
```
> mean(pred == churnTest$churn)
[1] 0.9634073
```

For reference, logistic regression with no interactions gives test accuracy of ~87% and logistic regression with all pairwise interactions gives test accuracy of ~78%

# A Complex Relationship



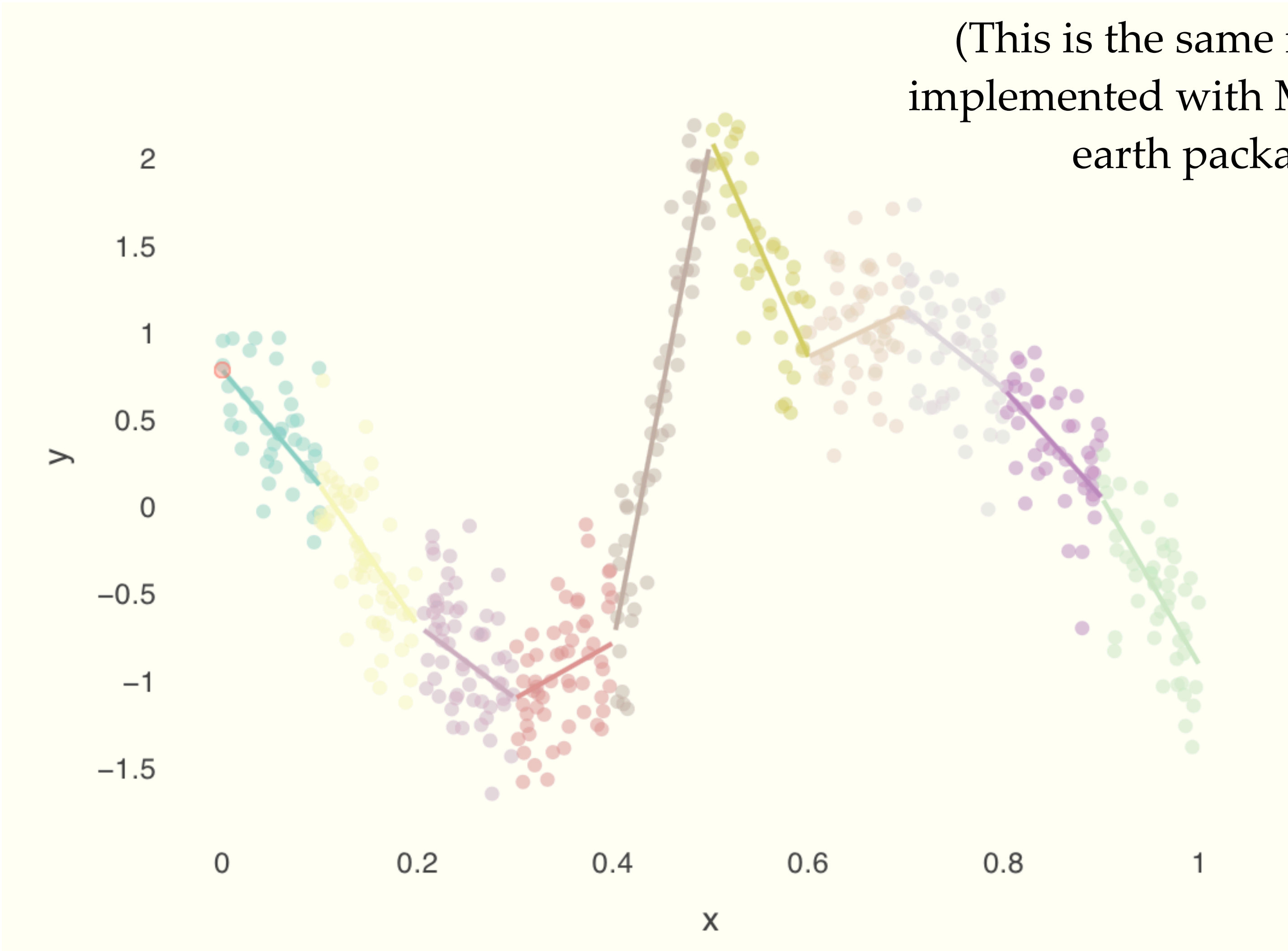
# Polynomial Regressions



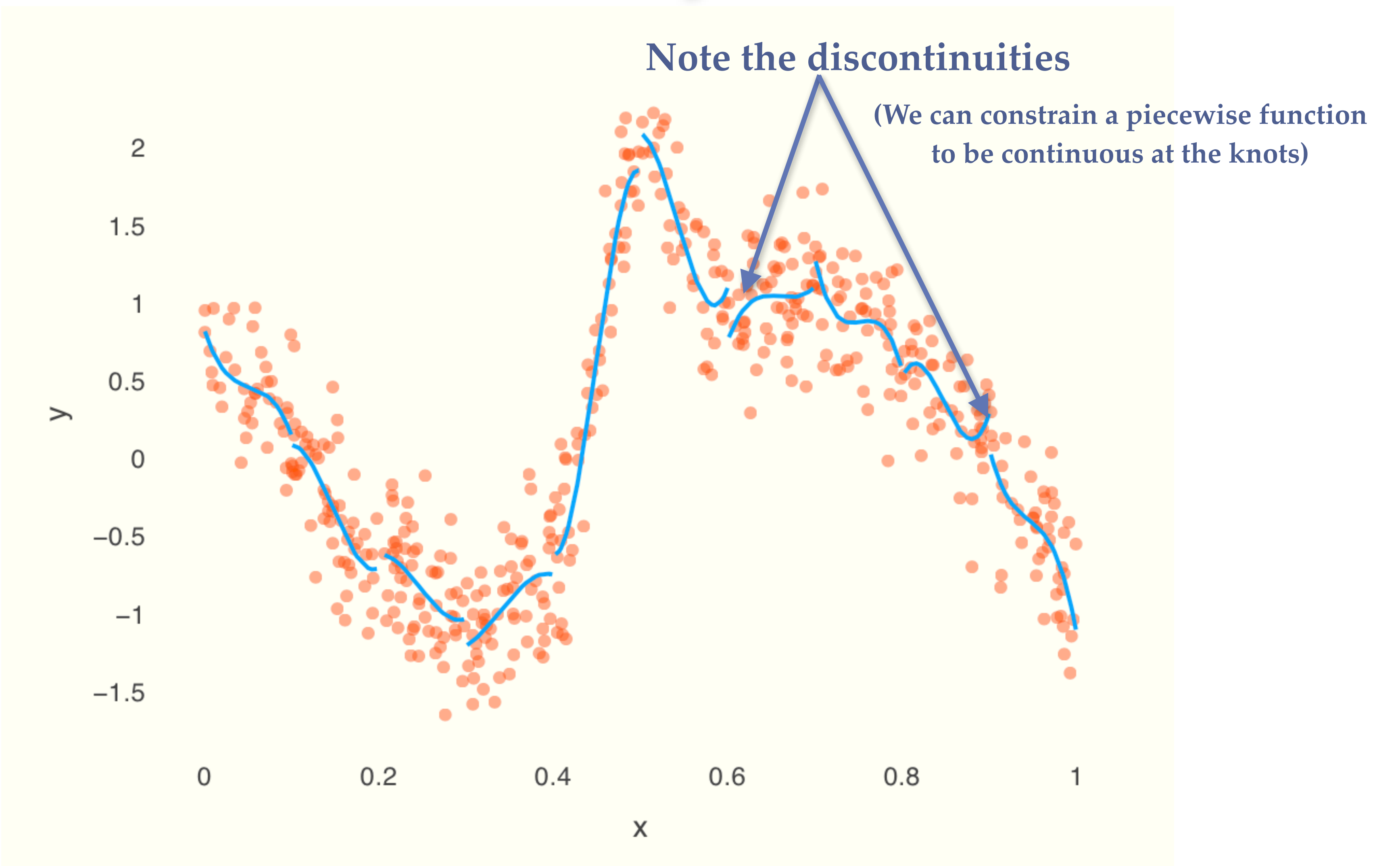
# Polynomial Splines

(degree = 1)

(This is the same model we implemented with MARS using earth package)



# Piecewise Polynomial Fit



# Cubic Splines

- In his original paper about MARS, Friedman suggests cubic functions to smooth out the jagged edges in the piecewise linear fits.
- This is not available in earth package but can be done in the base R package 'splines'

```
concrete$b_cement       =bs(concrete$cement      ,6)
concrete$b_slag         =bs(concrete$slag        ,6)
concrete$b_ash          =bs(concrete$ash         ,6)
concrete$b_water        =bs(concrete$water       ,6)
concrete$b_superplastic =bs(concrete$superplastic,6)
concrete$b_coarseagg    =bs(concrete$coarseagg   ,6)
concrete$b_fineagg      =bs(concrete$fineagg     ,6)
concrete$b_age          =bs(concrete$age         ,6)

cubic1 = lm(strength ~ b_cement + b_slag + b_ash + b_water + b_superplastic +
            b_coarseagg + b_fineagg + b_age, data = concrete)
```

Create Hinge Functions

Put into Linear Model

# Summary of Generalized Additive Models

**Advantages**:

- Allows a nonlinear relationship without trying out many transformations manually
- Often makes for improved predictions
- Inference still possible: we can still examine effect of each x on the target y while holding all other variables constant because model is additive.
- Fast!

**Disadvantages**:

- Can miss important interactions - although these can be incorporated into the model manually.

# Summary of Generalized Additive Models

**Notes:**

- Not the type of modeling technique that you can just throw in hundreds of variables - each variable should be prescreened with the target.
- GAMs not immune to multicollinearity.
- Picking smoothing parameters for each variable (or the number of knots) can be tedious